

Mohit Singh

☎ (+91) 6239120076 | ✉ mohitssr11@gmail.com | [🌐 LinkedIn](#) | [🐙 GitHub](#) | [🌐 Portfolio](#)

SUMMARY

Third-year Information Technology student with strong expertise in C++, data structures, algorithms, and OOP. Experienced in building high-performance full-stack web applications, real-time communication systems, and automated background sync engines. Skilled in database design, web performance optimization, and integrating intelligent Generative AI workflows with semantic search.

EDUCATION

National Institute of Technology, Jalandhar
B. Tech in Information Technology (CGPA: 8.27)

Aug. 2024 – May 2028 (Expected)
Jalandhar, India

Mukat Public School
XII (CBSE) – 92.4%

2023
Rajpura, India

ACADEMIC PROJECTS

Scout – Full-Stack Website & API Monitoring Platform | [Live Demo](#)

- Built a real-time monitoring engine using **Next.js** and **PostgreSQL** to track system uptime, precision latency, and network timeouts.
- Configured secure background **cron jobs** to execute automated, high-frequency pings across multiple web endpoints on a continuous schedule.
- Integrated **Recharts & AI diagnostics pipeline** to automatically analyze server error logs.

MeetNow – WebRTC Video Conferencing Platform | [Live Demo](#)

- Engineered P2P video/audio streaming using **WebRTC** and **Socket.io** to handle complex signaling, ICE candidates, and real-time screen sharing.
- Implemented secure session-based authentication and custom **Higher-Order Components (HOC)** for protected routing and persistent user states.
- Developed **synthetic media fallbacks** (Black/Silence tracks) and real-time chat with **MongoDB**-integrated history to ensure connection stability.

TECHNICAL SKILLS

Programming Languages : C, C++, SQL, JavaScript, TypeScript

Data Structures & Algorithms : Solved 500+ problems on LeetCode

Web Technologies & Frameworks : HTML, CSS, Next.js, React.js, Tailwind CSS, Bootstrap

Backend & Databases : Node.js, Express.js, Prisma ORM, MongoDB, PostgreSQL

Generative AI : LangChain, OpenAI & Gemini APIs, RAG

Core CS : OOP, DBMS